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LETTER TO THE EDITOR

Bilateral optic disc edema in a case of pycnodysostosis

Œdème papillaire bilatéral dans un cas de pycnodysostosis

Introduction

Pycnodysostosis is an inherited sclerosing bone abnormality characterized by short stature; prominent forehead and eyes with delayed cranial suture closure; a small face with a beaked nose, wide mandibular angle, a highly arched palate, and dental abnormalities; short and stubby fingers; increased bone density, osteosclerosis, bone fragility with multiple and recurrent fractures [1,2].

This disorder is caused by either homozygous or heterozygous mutations in the gene localized to 1q21 that encodes cathepsin K, a lysosomal cysteine protease that is highly localized in osteoclasts.

Here, we present a case of bilateral optic disc edema in a 10-year-old girl with a diagnosis of pycnodysostosis with typical features.

Case

In 2010, a 10-year-old girl was referred to neuro-ophthalmology clinic with decreased vision and bilateral proptosis without headache, tinnitus, diplopia. Her best corrected visual acuity (BCVA) was 20/40 Snellen units bilaterally, biomicroscopy was normal, fundus examination showed bilateral optic disc edema (Fig. 1a and b).

Peripapillary retinal nerve fiber layer thickness (RNFL) with optical coherence tomography was 92 μm in the right eye and 89 μm in the left eye (Fig. 2a).

Standard automated perimetry (HFATM II; Humphrey Instruments Inc., San Leandro, California, USA) with 30-2 visual field (VF) testing showed central and paracentral scotoma with blind spot enlargement in the right eye and paracentral and superior arcuate defects in the left eye (Fig. 3a).

External evaluation revealed prominent proptosis without lagophthalmia; there was no restriction in ocular motility.

On physical examination, she had a short stature (21 kg — 123 cm), prominent forehead, increased mandibular angle, irregular dentition (Fig. 4). Medical history revealed that she was diagnosed with pycnodysostosis at the age of 3 by a pediatric endocrinology clinic and in DNA strand analysis, a mutation in the 6th exone of cathepsin-K gene, located on the 1q21 locus, was detected. She was homozygous for the 1249 T mutation, which causes pycnodysostosis. Her parents were heterozygous for this mutation; consanguineous marriage was noted between them.

Computed tomography (CT) of the orbit revealed bilateral enlargement of optic nerve sheath (Fig. 5a). Cranial magnetic resonance imaging (MRI) showed no space-occupying lesion. MRI venogram was normal, B-scan ultrasound showed no drusen. The opening pressure in the lumbar puncture (LP) was found to be normal (150 mmH2O).

After increased intracranial pressure was ruled out, compression of the optic nerves in the optic canal was investigated. The patient was referred to a pediatric

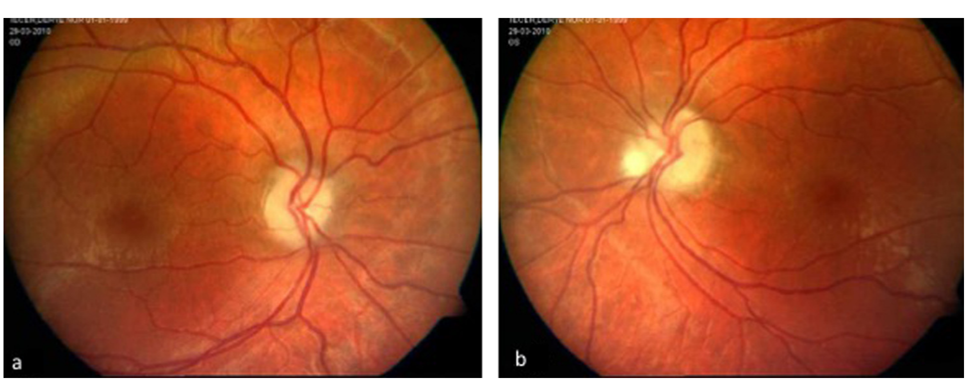

Figure 1. a: indistinct margins particularly at inferior, superior and nasal quadrants in the right eye; b: at first presentation, indistinct margins at all quadrant, accompanied with myelinated nerve fibers at nasal quadrant in the left eye.

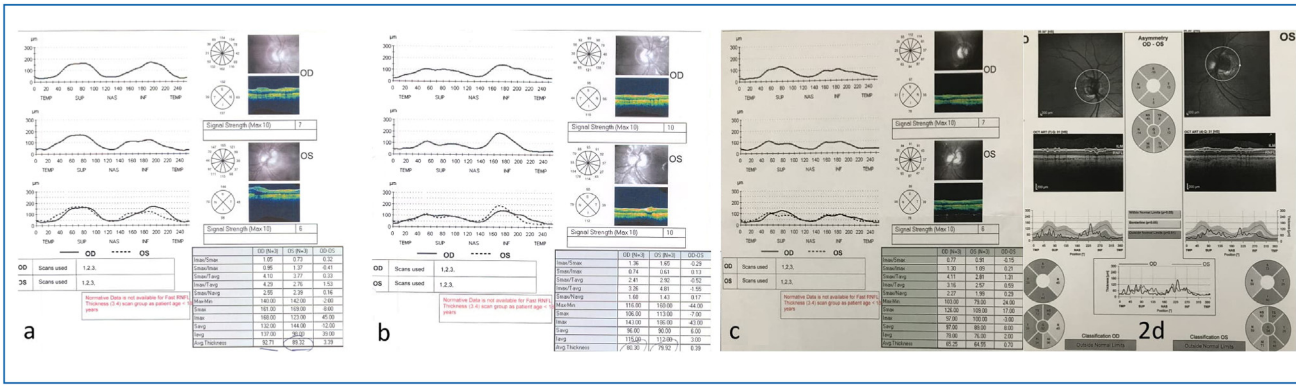

Figure 2. a: peripapillary retinal nerve fiber layer thickness with optical coherence tomography at the time of diagnosis; b: peripapillary retinal nerve fiber layer thickness with optical coherence tomography 3 months after the first decompression surgery; c: peripapillary retinal nerve fiber layer thickness with optical coherence tomography at the fourth year of follow up; d: peripapillary retinal nerve fiber layer thickness with optical coherence tomography 3 months after the second decompression surgery.

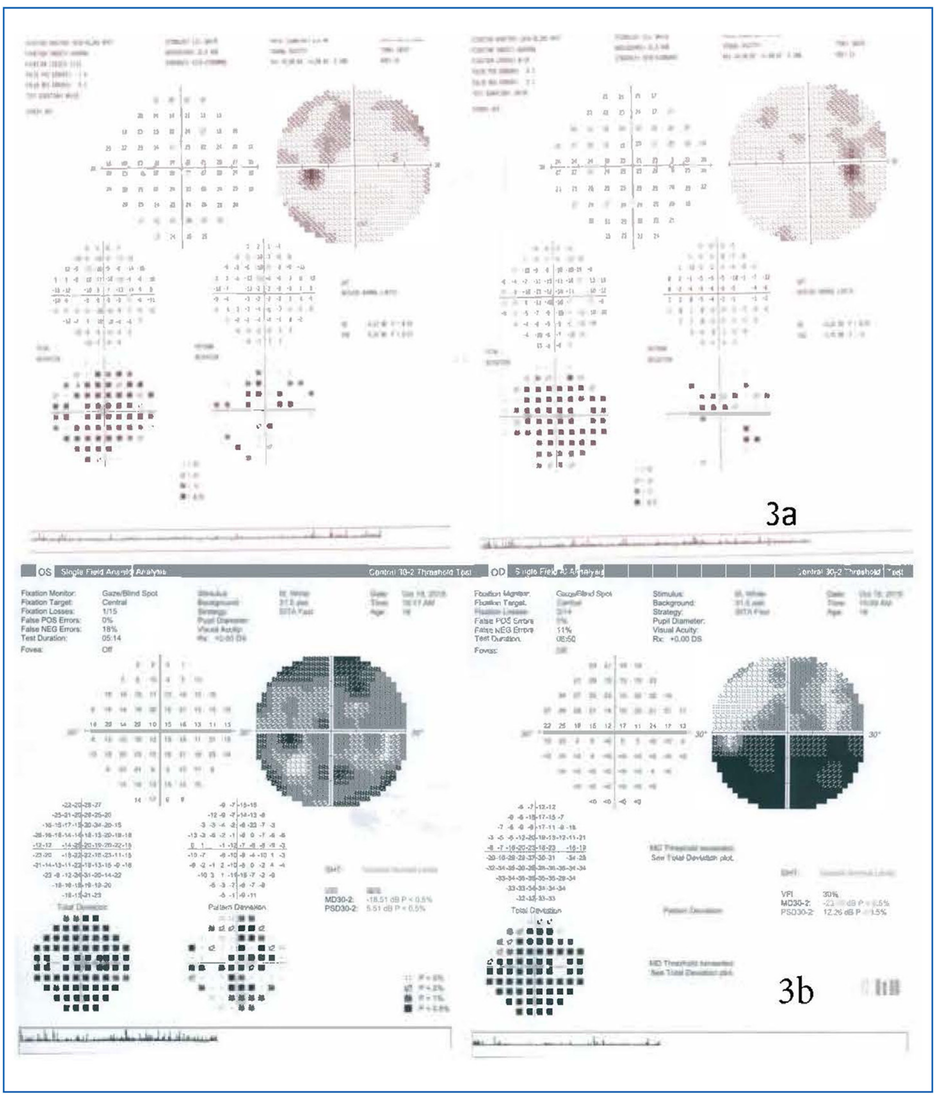

Figure 3. a: central and paracentral scotoma with blind spot enlargement in the right eye and paracentral and superior arcuate defects in the left eye; b: visual field defects at final examination. Inferior altitudinal defect in the right eye and general depression with central scotoma with superior and inferior arcuate defects in the left eye.

Figure 4. (a) External evaluation with prominent proptosis and prominent forehead. (b) External evaluation with irregular dentition.

neurosurgery clinic, where she underwent bilateral orbital unroofing and bilateral optic canal decompression.

Three months after surgery, BCVA was 20/30 Snellen units in the right eye and 20/40 Snellen units in the left eye. Funduscopic examination showed regression of the optic disc with decrease of RNFL thickness (Fig. 2b). Visual field analysis revealed an improvement in the defects.

Three years, her visual acuity remained stable in both eyes.

At the fourth year of follow-up, BCVA had deteriorated to 20/100 Snellen units bilaterally. In funduscopic examination, indistinct optic disc margins with pale nerve heads and atrophic RNFL were detected in both eyes (Fig. 2c).

In the presence of decreased visual function and VF progression, the patient was referred to the neurosurgery department for evaluation of the recurrence of optic canal narrowing. As repeated CT scans showed bilateral narrow optic canals (Fig. 5b), bilateral optic canal decompression surgery was performed for the second time.

Three months after the second decompression surgery, BCVA was 20/100 Snellen units in both eyes. Bilateral optic disc edema regressed with peripapillary RNFL thicknesses of 52 μm in the right eye and 54 μm in the left eye (Fig. 2d); VF improvement was detected.

Despite all the surgical interventions, optic nerve compression continued during follow-up and the neurosurgery department did not recommend anymore optic canal or orbital decompression surgeries. At the final examination, 9 years after the first referral to our clinic, BCVA was 20/100 Snellen units bilaterally. Fundus examination revealed atrophy of the optic discs with RNFL thicknesses of 48 μm in the right eye and 51 μm in the left eye. The Table 1 shows the correlation between RNFL and BCVA during the follow up. VF showed an inferior altitudinal defect in the right eye and central scotoma with superior and inferior arcuate defects in the left eye (Fig. 3b).

Narrow optic canals with bony regrowth were shown in repeated CT scans (Fig. 5c).

Discussion

Pycnodysostosis is an autosomal recessive disease that was first described by Maroteaux and Lamy in 1962 [1]. The prevalence is 1 to 1.7 per million and is equally distributed in both sexes [2]. Cathepsin K, a lysosomal enzyme secreted by osteoclasts that allows the division of proteins of the bone matrix, has an essential role in bone resorption and remodeling. The abnormal bone turnover due to cathepsin K deficiency leads to osteosclerosis and a bone structure that

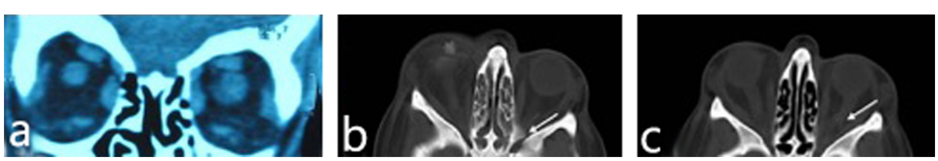

Figure 5. a: coronal section CT scan of the orbit revealed bilateral enlargement of the optic nerve sheath; b: bone window axial CT scans. Bilateral narrow optic canals particularly in the left eye (white arrow) before second optic canal decompression surgery; c: bone window axial CT scans. Bilateral narrow optic canals with bony regrowth and enlarged optic nerve sheat with ''S'' curved shape (white arrow) at final examination.

Table 1 Correlation between retinal nerve fiber layer thickness and visual acuity during the follow up.

	Right eye		Left eye	
	RNFL (μm)	BCVA	RNFL (μm)	BCVA
At the time of diagnosis	92	20/40	89	20/40
Three months after the first decompression surgery	80	20/30	79	20/40
At the fourth year of follow-up	65	20/100	64	20/100
Three months after the second decompression surgery	52	20/100	54	20/100
Nine years after the first diagnosis	48	20/100	51	20/100

RNFL: retinal nerve fiber layer; BCVA: best corrected visual acuity.

cannot adapt to mechanical stress; this causes an increased susceptibility to fractures [3,4].

Elmore et al. described the features of pycnodysostosis in 33 patients with short stature, increased bone density on X-ray, increased mandibular angle, proptosis, increased bone fragility, short dysplastic distal phalanges, open fontanelles, hypoplastic paranasal sinuses, dysplastic clavicles, hypoplastic facial bones, and consanguinity in parents [5].

As short stature is essential, many patients are referred to the pediatric unit. Radiological techniques to confirm osteosclerosis and acroosteolysis, combined with genetic analysis, are required for diagnosis.

At the age of 3, our patient was admitted to an endocrinology clinic with growth retardation. A genetical investigation was performed and homozygous 1249 T mutation that causes pycnodysostosis was found. Various novel mutations of cathepsin K gene in pycnodysostosis have been reported in the literature [6].

The patient was admitted to our clinic with the complaint of decreased vision and bilateral proptosis at 10 years of age with the features of pycnodysostosis.

Fundus examination revealed bilateral optic disc edema. In the literature, papilledema has been reported in two case reports due to increased intracranial pressure with craniosynostosis caused by pycnodysostosis [7,8].

In our patient, since the LP opening pressure was normal, papilledema was ruled out. On CT scans, bilateral shallow orbital fossa and dilatation of the optic nerve sheath suggested optic nerve compression in the optic canal and optic canal decompression surgery was performed. In the literature, it was claimed that cranial nerve compression was not a feature of pycnodysostosis [9]; however, our patient had a genetical proof of pycnodysostosis. After the orbital decompression surgery, a good visual outcome was achieved.

In the literature, there are reported cases of successful orbital decompression surgeries with good visual outcomes in patients with osteopetrosis [10]. The differential diagnosis of pycnodysostosis includes osteopetrosis, cleidocranial dysplasia and idiopathic acroosteolysis. In osteopetrosis, the bone marrow may be absent and hematopoietic alterations are common. Signs of compression of the cranial nerves such as facial paralysis, deafness or pain may exist. Cranial dysplasia may seem like pycnodysostosis however, bone density is not increased. In idiopathic acroosteolysis, the appearance of the patients is typical, with hypotelorism, exophthalmos and an upturned nose. The angle of mandible is acute and increased bone density is not present [11] To our knowledge, ours is the first case of bilateral optic disc edema due to optic nerve compression accompanied by proptosis in a child with pycnodysostosis. According to us, the cause of this compression and proptosis was shallow orbital fossa and progressive osteosclerosis. In pycnodysostosis, we have to consider optic nerve compression. Regular ophthalmologic examination with visual function tests, funduscopic examination, VF analysis, and color fundus photography should be performed at the time of diagnosis and at follow-up. We recommend that these patients should be offered to the ophthalmologist before their visual acuity decreases.

Conclusion

This study is the first to report a rare case of pycnodysostosis with bilateral optic disc edema due to optic nerve compression. Ophthalmologic examination and orbital radiological techniques played key roles in the definitive diagnosis.

Ethical statement

This manuscript has not been published and is not under consideration for publication elsewhere.

Disclosure of interest

The authors declare that they have no competing interest.

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